



M000 Week 0 — English Worksheet

Topic: Pixel Art — Counting Squares · **Course:** Builder Basics · **Type:** Warm-Up (before Week 1) ·

Time: about 40 minutes

Every block in Minecraft has three numbers: **X** (left ↔ right), **Y** (up ↓ down), and **Z** (depth, in ↔ out). Pixel art sits on a flat wall, so **Z stays the same** the whole time — you only change **X** and **Y**.



This is a quick warm-up to do **before Week 1**. It teaches you to count squares so your builds line up perfectly.

1 · Read the Wall 🔍

Open the **Pixel Art** world. In front of you is a flat wall with red pixel art on it.

Think of the wall like graph paper:

- **X** counts the columns, left to right →
- **Y** counts the rows, bottom to top ↑
- **Z** is how deep the wall is. It is the **same** for every red square, so we can ignore it here.

The big idea for this whole worksheet:



To color **one** square, your **start** and your **end** are the **same** square. A 1-wide, 1-tall block goes **from (X, Y) to the same (X, Y)**.

Look at the wall and find the smallest red mark — a single square. In your own words, why are its start and its end the same?

2 · One Square: Start = End

Here is a grid. **X** runs across the top, **Y** runs up the side. One square is red.

Y \ X	0	1	2	3	4	5
3						
2						
1						
0						

The red square is at **X = 3, Y = 2**. Here is the same command two ways — in **blocks** and in **python**:

Blocks

```
fill [  red wool ]  
  from world X 3 Y 2 Z 0  
  to   world X 3 Y 2 Z 0
```

Python

```
blocks.fill(RED_WOOL, world(3, 2, 0), world(3, 2, 0), FillOperation.Replace)
```

Look closely: the **from** and the **to** are **exactly the same** — `(3, 2, 0)` and `(3, 2, 0)`. One square in, one square colored.

💡 For a single block you can also use `blocks.place(RED_WOOL, world(3, 2, 0))` — place needs only **one** spot, because there is nothing to count.

A single block has the same start and end. So why does `fill` make you write the numbers twice instead of once?

3 · Predict 🧠

Read each command. Before you picture it, write how many squares turn red and where.

```
blocks.fill(RED_WOOL, world(2, 1, 0), world(2, 1, 0), FillOperation.Replace)
```

How many squares turn red? Where?


```
blocks.fill(RED_WOOL, world(2, 1, 0), world(5, 1, 0), FillOperation.Replace)
```

Y stays 1. X goes 2 → 5. How many squares turn red? List every X number it covers.


```
blocks.fill(RED_WOOL, world(4, 0, 0), world(4, 3, 0), FillOperation.Replace)
```

X stays 4. Y goes 0 → 3. Is this line flat or standing up? How tall is it?

4 · Count the Length 12 34

Counting squares has a trap. From $X = 2$ to $X = 5$ is **not** 5 squares — it is **2, 3, 4, 5 = 4 squares**. You count the start square too.

Length = (end – start) + 1. So to make a line **N** long starting at X , the end is $X + (N - 1)$ — not $X + N$.

You want a flat red line that is 3 squares long, starting at $X = 1$ (Y stays 2). What is the end X ?

Now write the fill for it — blocks or python, your choice:

You want a line standing up, 5 squares tall, starting at $Y = 0$ (X stays 6). What is the end Y ?

5 · Spot the Bug 🐛

Each command was meant to do one thing, but it is wrong. Rewrite it so it works, then explain why the original was wrong.

Bug A — Meant to color **one** square at X 4, Y 4.

```
blocks.fill(RED_WOOL, world(4, 4, 0), world(5, 5, 0), FillOperation.Replace)
```

Hint: count how many squares (4,4) to (5,5) really covers.

Write the fixed command:

Why was it wrong? Why does your fix work?

Bug B — Meant to make a flat line **3** squares long starting at X 0 (Y stays 1).

```
blocks.fill(RED_WOOL, world(0, 1, 0), world(3, 1, 0), FillOperation.Replace)
```

Write the fixed command:

Why was it wrong? Why does your fix work?

Bug C — Meant to make a line **standing up**, 4 tall at X 2, starting Y 0.

```
blocks.fill(RED_WOOL, world(2, 0, 0), world(5, 0, 0), FillOperation.Replace)
```

Hint: for a standing line, which number should change — X or Y?

Write the fixed command:

Why was it wrong? Why does your fix work?

6 · Build It 📷

Now switch to your homework world. Recreate the red wall. Start with the single squares (start = end), then the flat lines, then the standing lines. Do as much of the wall as you like.

Send a photo or video of your wall, then explain what you did. Use these sentence starters — write 4 to 6 sentences total.

The first square I colored was at X ... , Y ...

For one square, my start and end were the same because ...

To make a line ... long, I counted ...

The hardest part was ...

To fix it, I ...

Next time I would ...

7 · Record Your Walkthrough

Take a video on your phone (or a parent's phone) while you build one square and one line. Talk through it like you are teaching someone who has never seen it. Try to use these words: **X, Y, start, end, same, fill.**

1. Show the empty wall and point out X (across) and Y (up).
2. Color one square and say why the start and the end are the same.
3. Make a line and count the squares out loud.
4. Show one counting mistake and how you fixed it.

Write what you will say in your video. Plan it below before you record — you can read from it while filming.

Submit 

Send this worksheet + your walkthrough video to teacher on KakaoTalk.